

Ch. 7 — P04 (K) M/G/1 and the Effect of Service-Time Variance

Problem Bank | Chapter 7 | Type: K | Difficulty:

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Objective

Demonstrate empirically that the mean waiting time in an M/G/1 queue depends on the *variance* of service time — not just its mean — and verify the Pollaczek–Khinchine (P-K) formula against SimPy simulation.

Background

The P-K mean waiting-time formula for an M/G/1 queue is:

$$W_q = \frac{\lambda \mathbb{E}[S^2]}{2(1 - \rho)}$$

where $\rho = \lambda/\mu$, $\mathbb{E}[S] = 1/\mu$, and $\mathbb{E}[S^2] = \text{Var}(S) + (1/\mu)^2$.

Setup

Fix $\lambda = 3/\text{hour}$ and $\mu = 4/\text{hour}$ ($\rho = 0.75$). Consider four service-time distributions, each with the same mean $1/\mu = 0.25$ hr but different variance:

Distribution	Parameters	$\text{Var}(S)$	$c_s^2 = \mu^2 \text{Var}(S)$
Deterministic (D)	constant = 0.25 hr	0	0
Erlang-2 (E_2)	shape= 2, rate= 8/hr	$1/(2 \cdot 16)$	0.5
Exponential (M)	rate= 4/hr	$1/16$	1.0
Hyperexponential	see below	—	2.0

For the hyperexponential: with probability 0.5 draw from $\text{Exp}(\text{rate}= 2/\text{hr})$; with probability 0.5 draw from $\text{Exp}(\text{rate}= 6/\text{hr})$. Verify that this gives mean 0.25 hr and $c_s^2 = 2$.

Tasks

1. **Compute** the theoretical W_q for each distribution using the P-K formula.
2. **Implement** a general M/G/1 SimPy simulation:
3. **Run** 20 replications for each distribution and compute 95% CIs for W_q .
4. **Plot** simulated W_q (with error bars) and theoretical P-K W_q on a single bar chart, grouped by distribution. Use a color-blind-safe palette.
5. **Sensitivity sweep:** Fix the exponential distribution ($c_s^2 = 1$) and vary ρ from 0.1 to 0.95 in steps of 0.05. For each ρ , compute theoretical P-K W_q and simulate (10 reps each). Plot both curves on the same axes. At what ρ does simulation variance become so large that the CI no longer contains the theoretical value?

Submission

Submit a Jupyter notebook with all code, the comparison plot, the sensitivity-sweep plot, and a written interpretation (≤ 150 words) of why variance matters more at high utilization.